

Social Impacts of Infrastructure

ENGR49580D Technology, Society and Sustainability

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Social Impacts of Infrastructure

- Context
- Eminent Domain
- Parks
- Disaster Response
- Public Transit
- Freeway Construction
- Take Away Points

Context

- The term “equity deserving group” is used in this lecture to identify groups that by virtue of their circumstances or by the actions of others are marginalized or otherwise less able to participate in society
- These groups include (but are not limited to):
 - Persons with disabilities
 - Persons experiencing homelessness or poverty
 - First Nations
 - Racialized persons
 - Persons who are LGBTQ+
 - Women

**You may identify as a member of
one or more of the groups
discussed today or of other
groups not mentioned.**

The contents of this lecture are based in research and are not intended to reflect individual experiences.

Nothing said today is intended to invalidate your individual experiences.

Eminent Domain

Expropriation

- Eminent domain is a term to describe a number of legal means by which the government may take possession of privately held property for public use
 - This may include other forms of possession such as easements, rights of way and leases.
- In Canada this process is called expropriation and controlled by the Expropriation Act or various provincial acts
 - Generally these acts lay out a process that includes public consultation prior to expropriation

Eminent Domain

Compensation

- Under the various acts that enable expropriation, the former owner of the property is entitled to compensation that may include
 - Fair market value of the property
 - Appraisal costs
 - Legal costs
 - Value to the owner
 - Value of an equivalent residence
 - Moving costs

Eminent Domain

De Facto Expropriation

- Constructive taking (de facto expropriation) occurs when a government uses other legal means (such as zoning) to limit the owner's use of a property in a way that deprives them of reasonable economic uses
- In Canada, owners may have legal claim against governments for constructive taking

Eminent Domain

Expropriation for Economic Development

- Some jurisdictions allow land to be expropriated with the intent of transferring it to a private developer
- It may be necessary to demonstrate in this case that the property in question is blighted

Eminent Domain

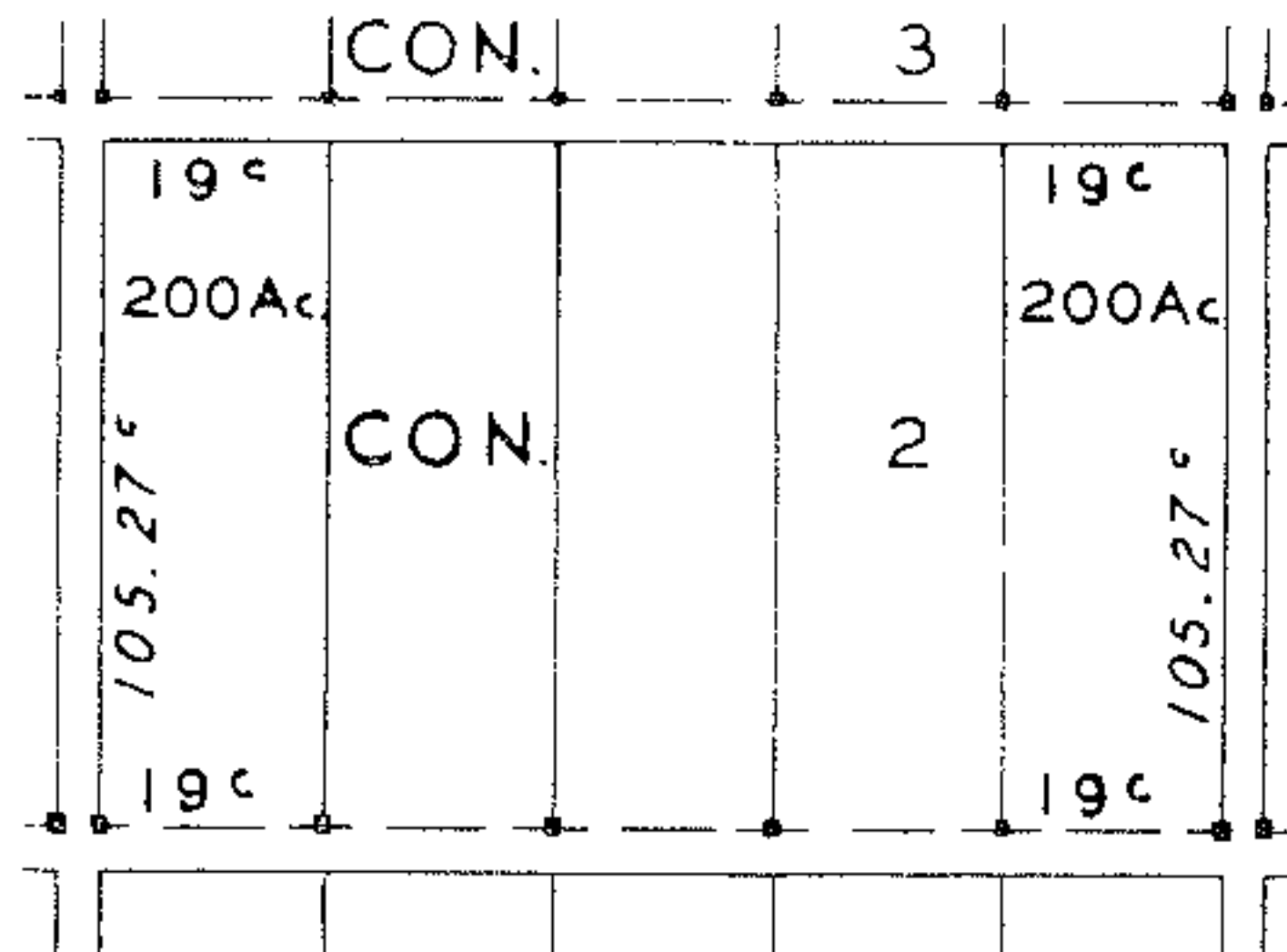
Other means of acquiring real-estate

- Open market purchases, private offers
- Setting aside land in plans of subdivision
- Land set aside on crown surveys



Typical section of a single-front township, 1783-1818

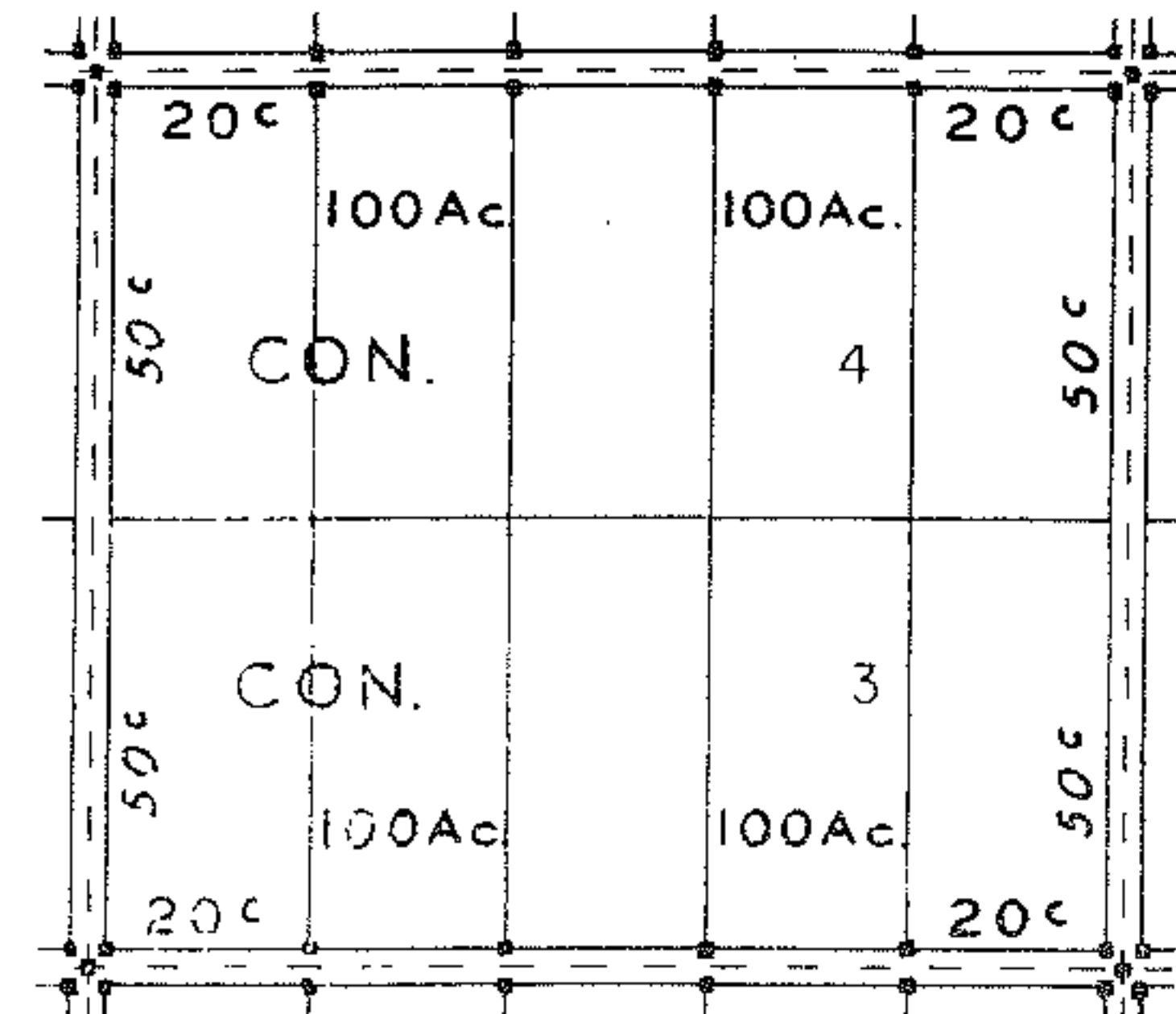
Original survey lines run on the ground are indicated by broken lines; original survey posts by small squares.



Typical section of 1,000-acre sectional system, 1835-1906

Sectional Township with Double Fronts

Original survey lines run on the ground are indicated by broken lines; original survey posts by small squares.



In Class Activity

- Think of a contemporary infrastructure project
 - Who does it benefit?
 - Who does it impact?
 - Who does it displace?

Think groups, not individuals

Parks

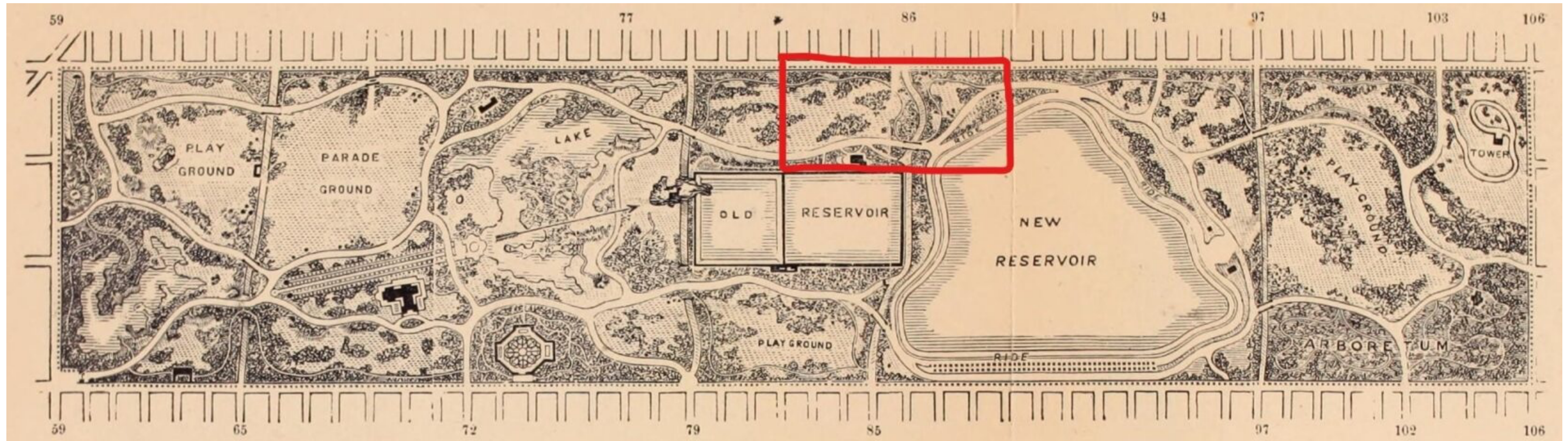
History

- Many 19th century parks were created to create buffer areas between elites and other groups
 - Used to perpetuate racial segregation
 - The park *improved* (increased the value of) urban real estate while excluding native and minority groups

Parks

Case Study: Central Park and Seneca Village

- The lands that make up central park were settled at the time of construction



Parks

Case Study: Central Park and Seneca Village

- Population of Seneca Village was 350+ people, ~2/3 Black and ~1/3 Irish
- Land was developed and owned by the residents
- Seneca and other villages to become the park were described by politicians and press as “shantytowns”
- Residents were called “denizens,” “squatters,” “vagabonds,” “scoundrels”

Seneca Village, New York City. National Park Service: ND. Available: <https://www.nps.gov/articles/seneca-village-new-york-city.htm>

Before Central Park: The Story of Seneca Village. Central Park Conservancy Magazine: 2018. Available: <https://www.centralparknyc.org/articles/seneca-village>

J. Reddick. Before Juneteenth: The Story of Seneca Village and Central Park. Central Park Conservancy Magazine: 2021.
Available: <https://www.centralparknyc.org/articles/before-juneteenth-the-story-of-seneca-village-and-central-park>

Parks

Case Study: Central Park and Seneca Village

- ~1600 residents displaced and lands seized by eminent domain in 1857
- While the residents were compensated under eminent domain, many considered their land undervalued in the process

Seneca Village, New York City. National Park Service: ND. Available: <https://www.nps.gov/articles/seneca-village-new-york-city.htm>

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Disaster Response

- Typical post disaster infrastructure restoration work prioritizes restoring the failure that affect the greatest number of customers without regard for the circumstances of those customers
 - Vulnerable groups may be more dependant on infrastructure or less able to afford backups such as generators or water storage
- Rural and remote areas have less infrastructure redundancy and are therefore more vulnerable

Public Transit

Effects on Equity Deserving Groups

- Transit helps enable low income households to access employment, educational opportunities and other services such as healthcare
- Land speculation and development associated with transit development can increase housing prices near transit
- Social housing development is rarely coupled with transit development

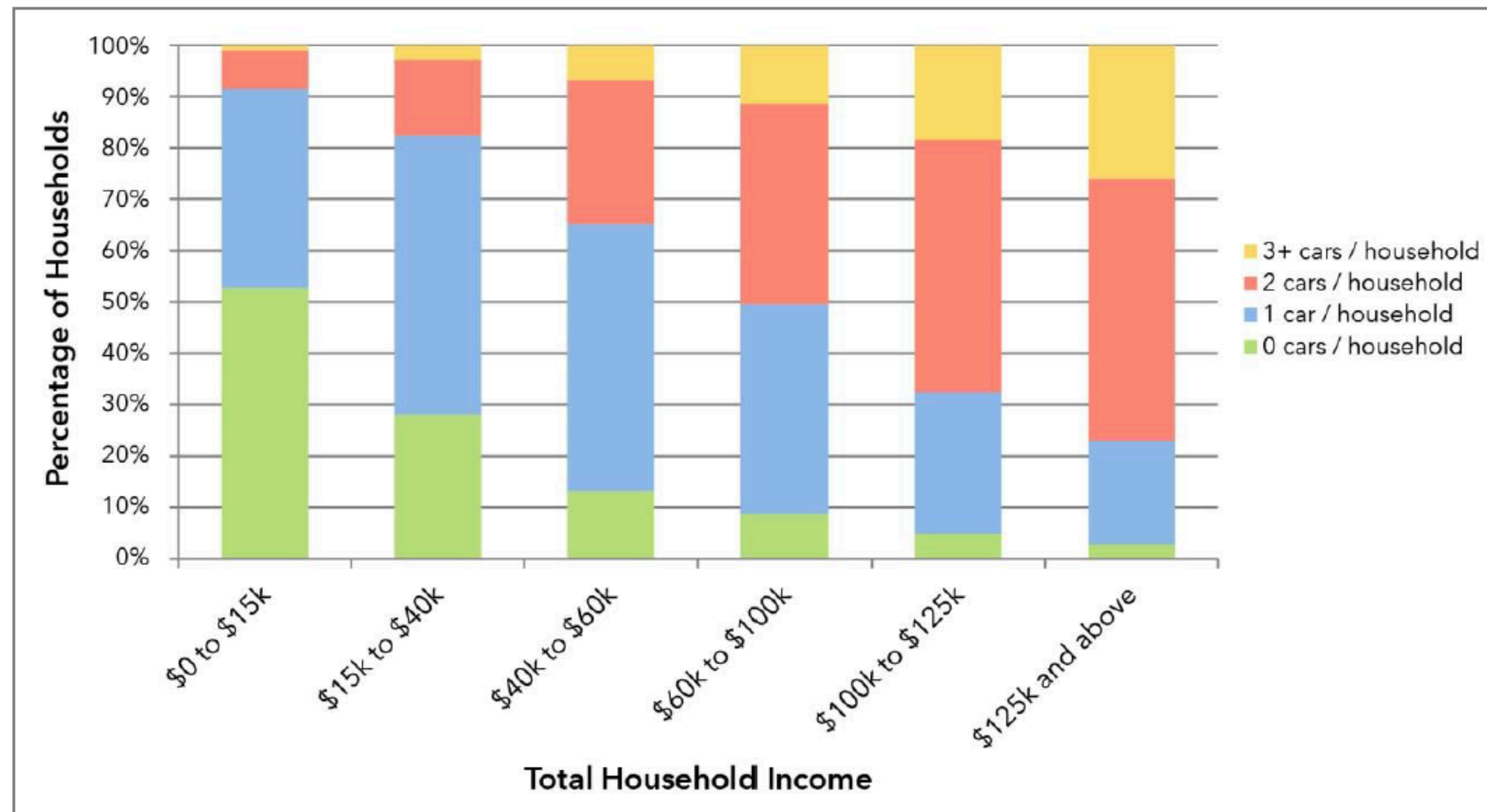
Public Transit

Reason for Non-automotive travel

- *People with disabilities (3-5% of residents)*
- *Seniors who do not or should not drive (5-15% of residents and increasing)*
- *Adolescents (5-15% of residents)*
- *Stay-at-home parents in single-vehicle household (varies)*
- *Low-income households (20-40% of households)*
- *Drivers who temporarily lack a vehicle (1-3%)*
- *Tourists and visitors (varies)*
- *Law-abiding drinkers (1-2%)*
- *People who want to walk and bike for health and enjoyment (20-60% of residents, plus dogs)*

Public Transit

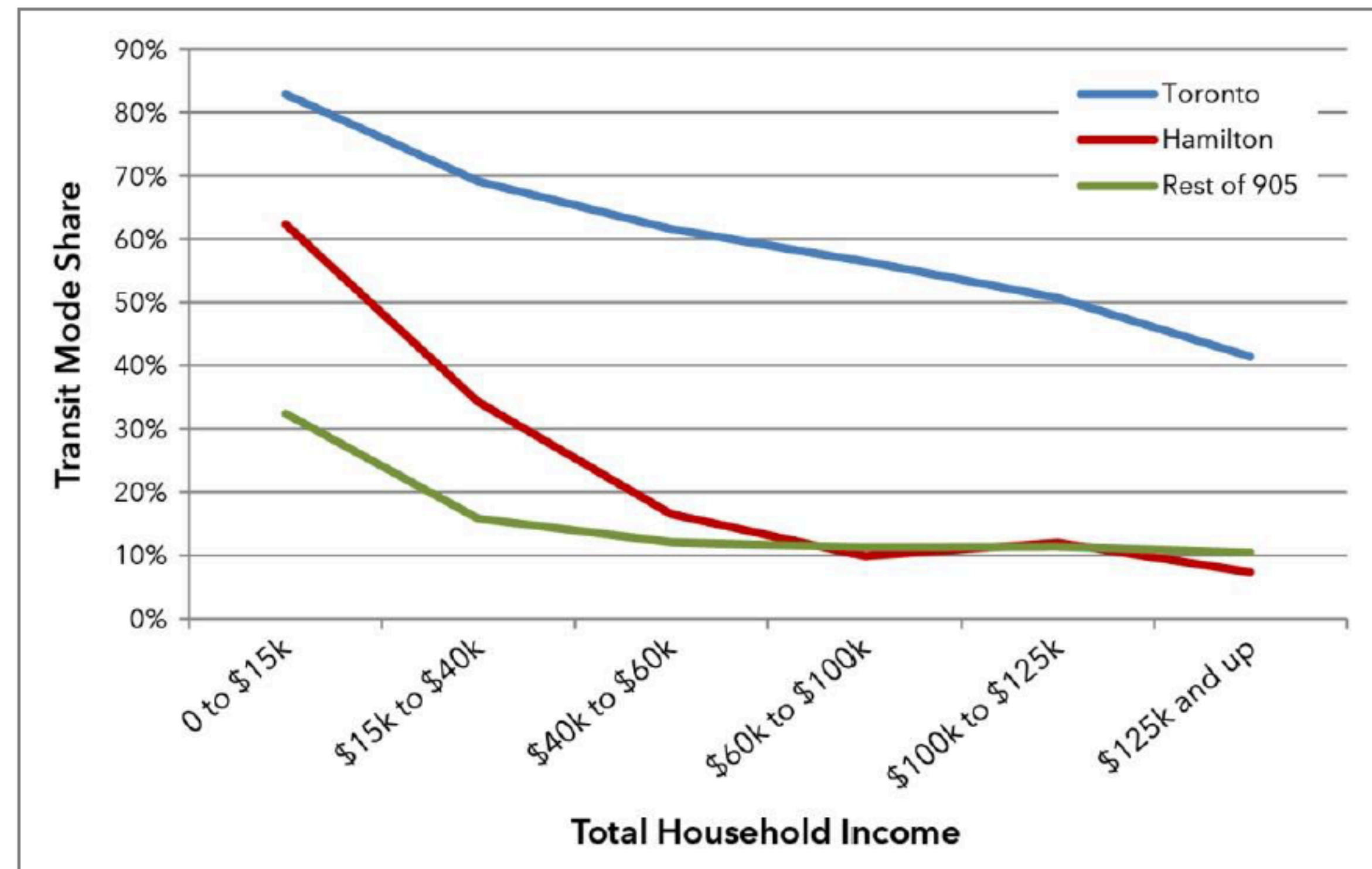
The Importance of Transit



Source: University of Toronto Data Management Group, 2016 Transportation Tomorrow Survey.

Public Transit

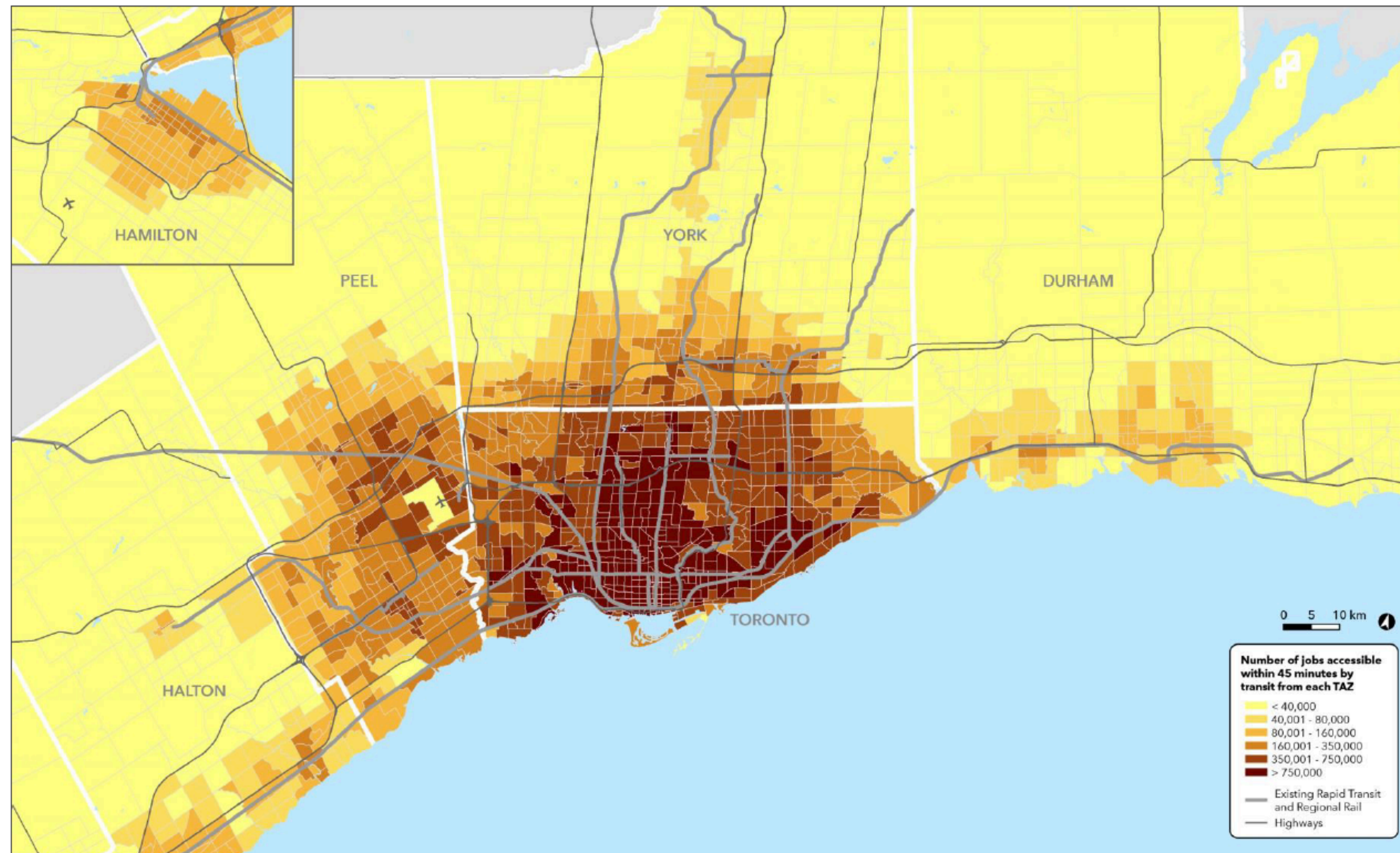
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Source: University of Toronto Data Management Group, 2016 Transportation Tomorrow Survey.

Public Transit

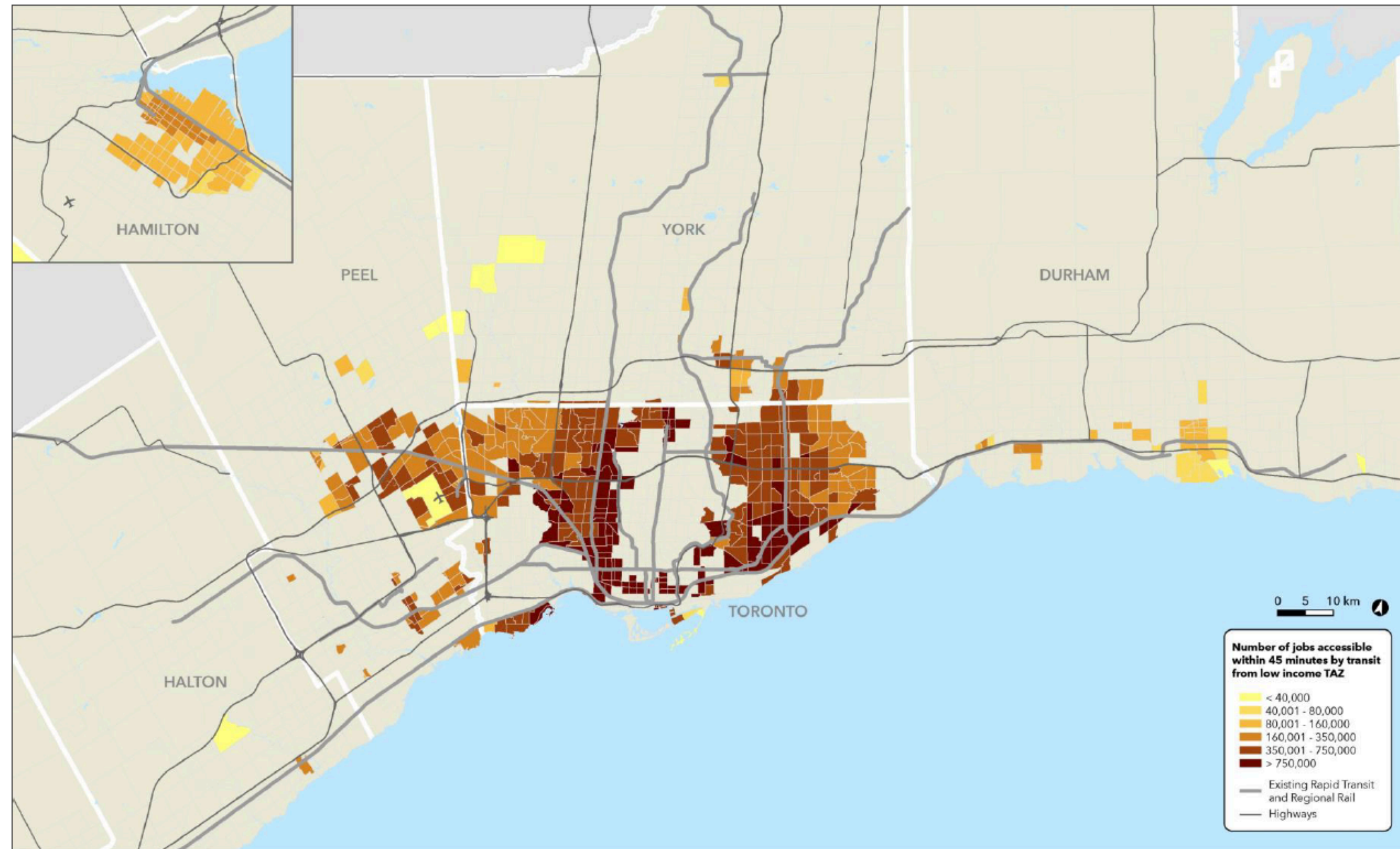
The Importance of Transit – Access to Jobs



Note: Data for Durham Transit were not available and transit scores in Durham Region reflect access to GO bus and rail only.

Public Transit

The Importance of Transit – Access to Jobs



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Metrolinx. Transit Access and Social Equity in the Greater Toronto and Hamilton Area. March 2018.
Available: https://assets.metrolinx.com/image/upload/v1663240133/Documents/Metrolinx/Transit_Access_and_Equity_2018.pdf

Public Transit

Effects on Equity Deserving Groups: Poor / Unavailable Service

- May impacted groups in urban; rural; on reserve and remote areas
- Access to destinations:
 - Employment; food; cultural services
 - Poor access to transit can result in individuals turning down employment
- Safety
 - Lack of transit drives vulnerable groups to more dangerous options

Public Transit

Effects on Equity Deserving Groups: Poor / Unavailable Service

- Long transit times:
 - Route design as well as transfers (combined with infrequent service) can increase the length of transit trips
 - Transit design may result in trips taking 4+ times the drive time

Public Transit

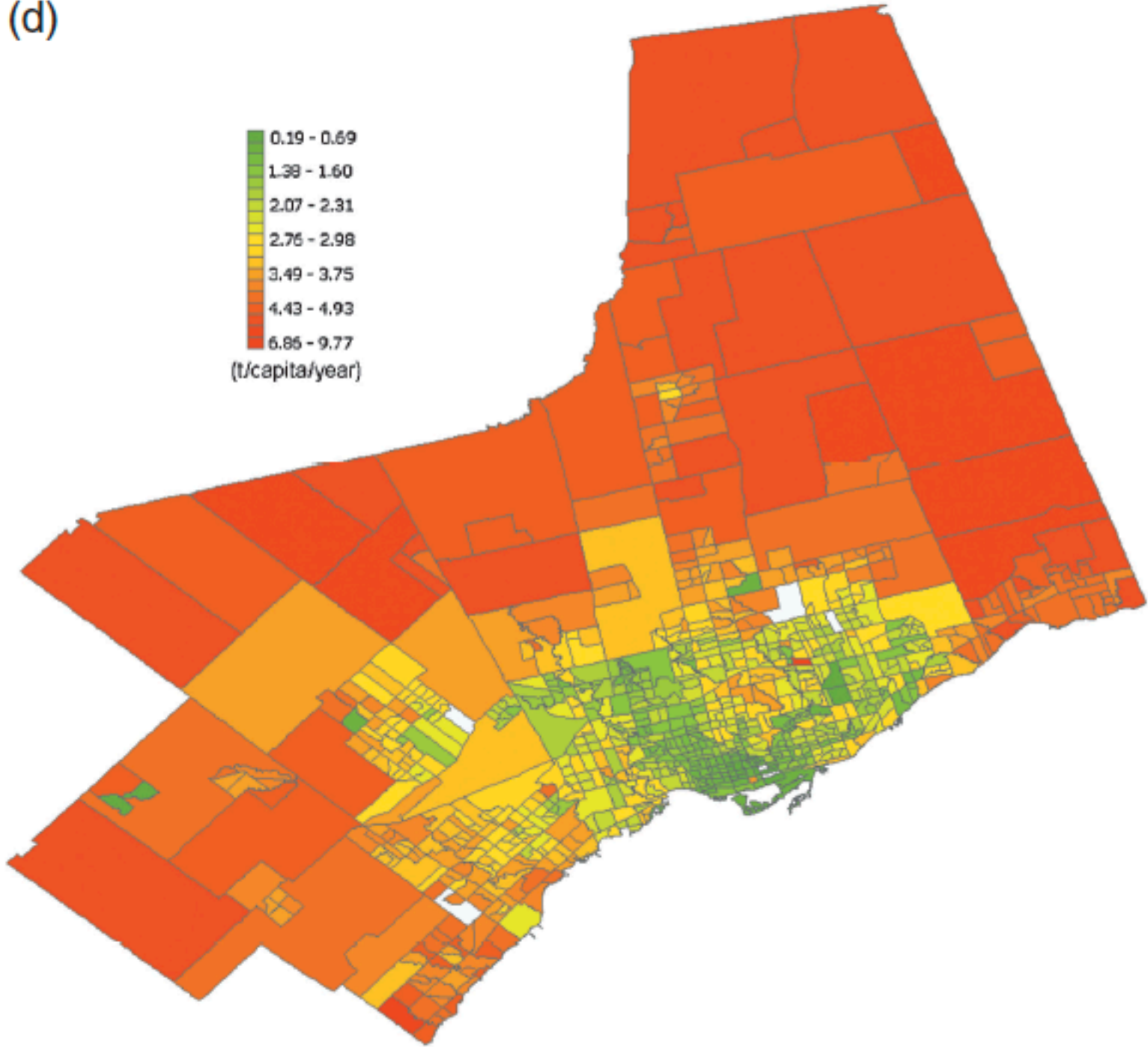
Effects on Equity Deserving Groups: Other Factors

- Unaffordability of Transit Fares
- Safety
- Policing and Enforcement

Public Transit

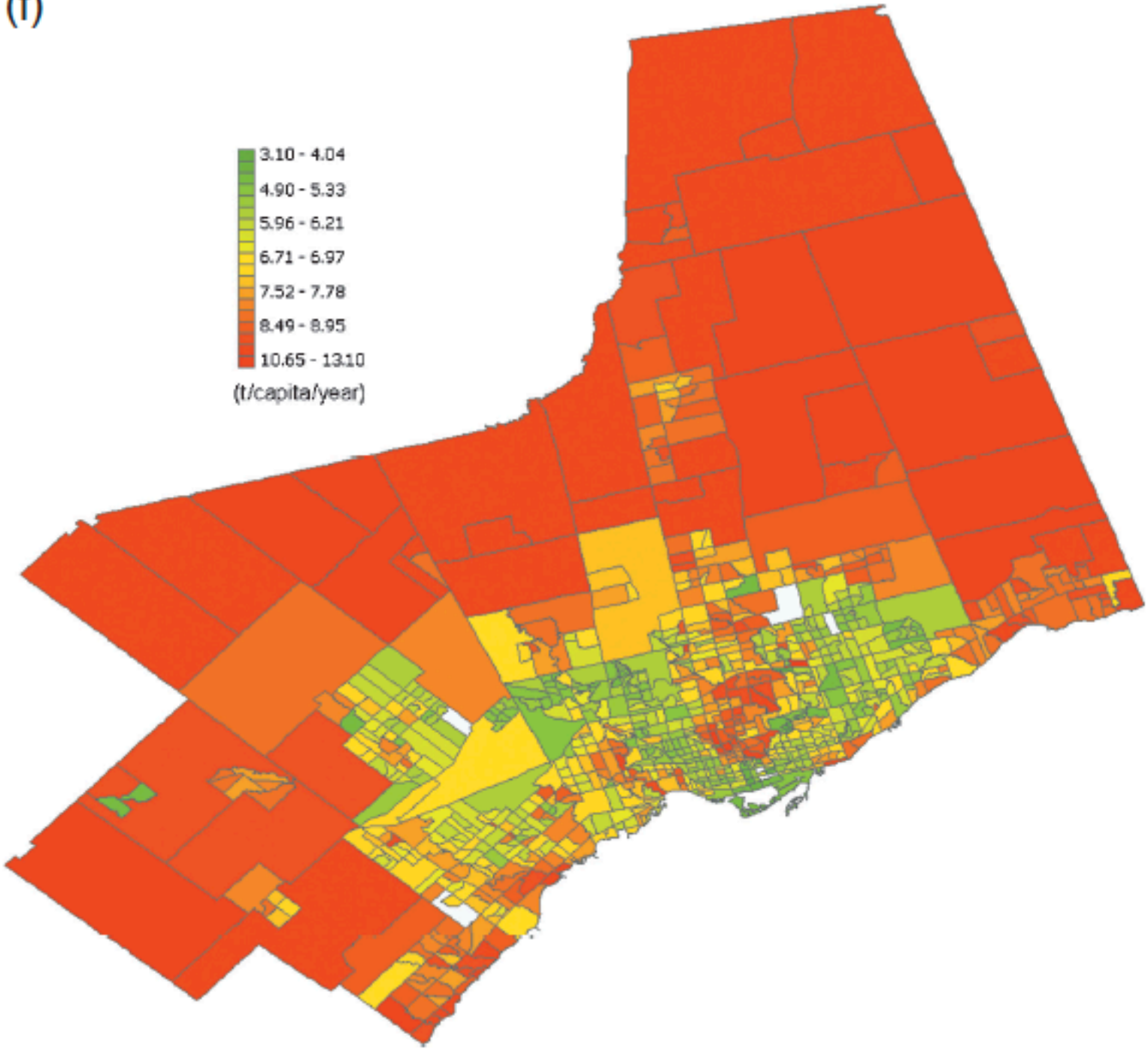
Climate Impacts

(d)



Transportation

(f)



Total Household

Freeway Construction

Impacts

- The construction of freeways requires the acquisition of large continuous swaths of land
 - This typically requires the use of eminent domain
- In the US more that 82 000 households (200 000 residents) were displaced due to highway construction between 1991 and 2021

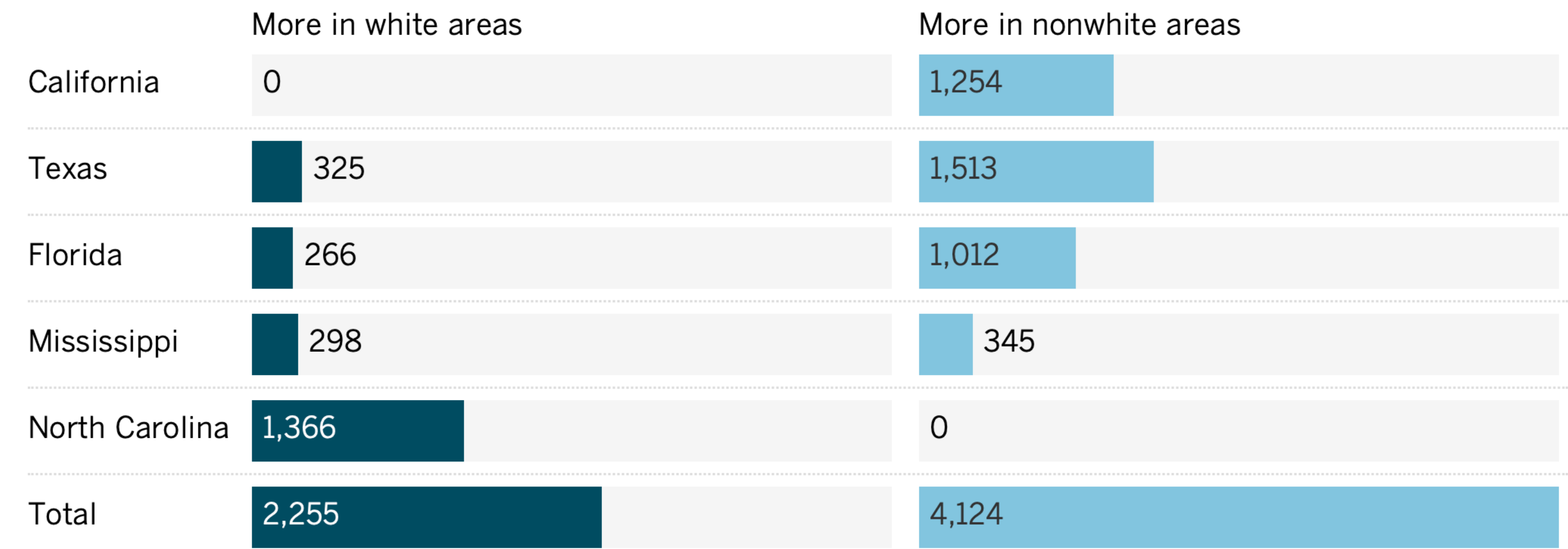
Freeway Construction

Impacts: Community Impact

Large highway projects leave uneven impact on communities of color

Nearly two-thirds of the homes taken for road projects analyzed by The Times occurred in those that disproportionately impacted Black and Latino areas over the past 30 years.

Homes taken ...



The Times analysis included projects it identified as taking more than 100 homes in the five states with the most displacements since 1991.

Federal Highway Administration; U.S. Census Bureau, state and local agencies

Freeway Construction

Impacts: Community Impact

- Beyond displacement, infrastructure construction including freeways can divide communities
- Blocking walking and other local transportation routes
- Separating residents from business and other services
- Devaluing real-estate



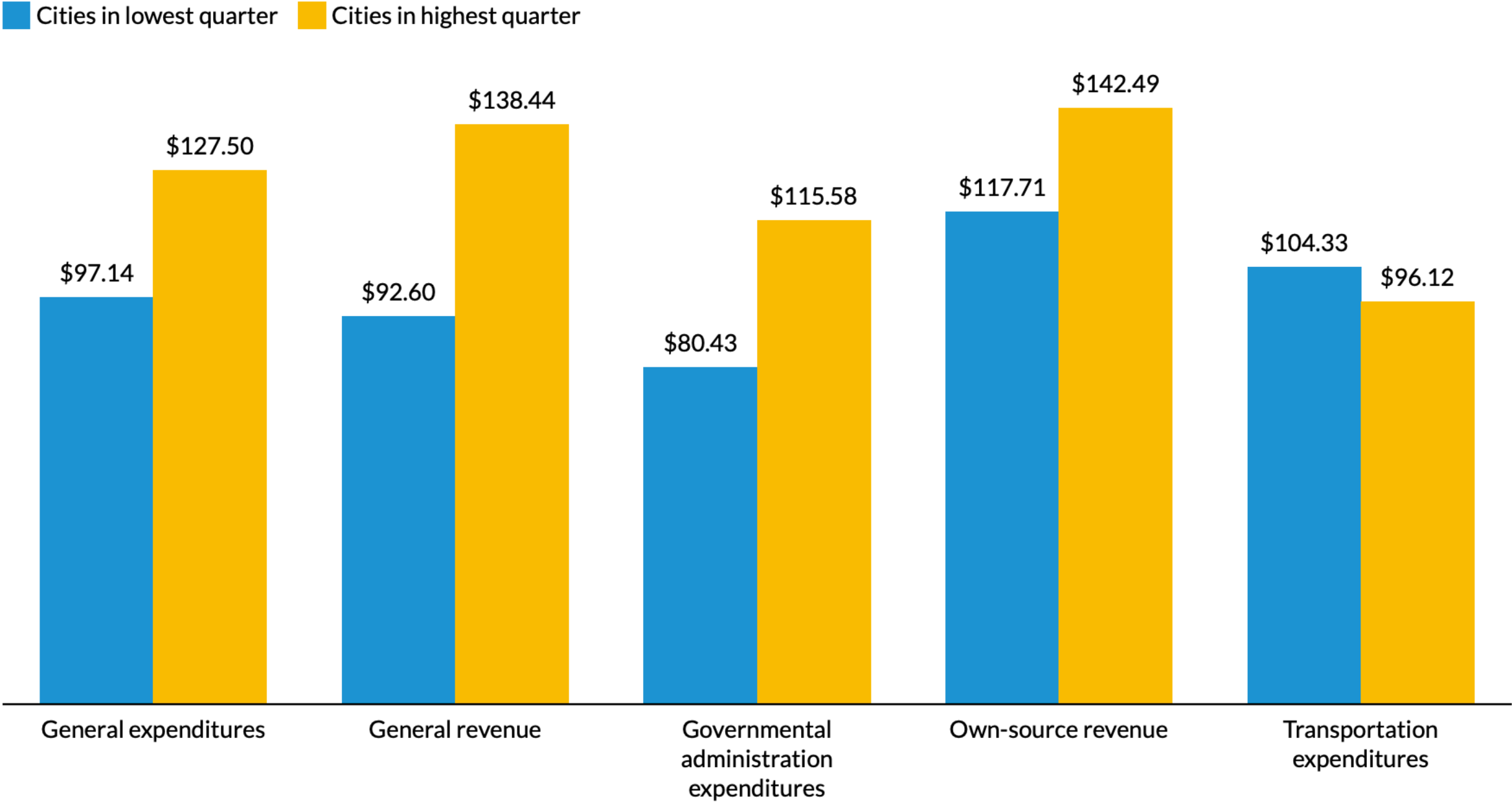
Take Away Points

Why infrastructure can have adverse effects on equity deserving groups

- Already marginalized communities lack the resources (political and financial) to oppose disruptive infrastructure projects (such as freeways)
 - These projects result in a loss of housing, business and connectivity within the community
- Infrastructure that benefits the affluent (such as highways) is often given priority over more inclusive infrastructure (such as transit)
- Marginalized communities are less able to maintain infrastructure leaving it vulnerable to disaster

Take Away Points

Municipalities with more resources are more likely to win grants



Take Away Points

How to effect change

- Engineers should consider (and bring forward) potential impacts of projects
 - This means doing more than designing to code
- Rely on community outreach and consultation
- Advocate for policy change
- Use Human Development Index metrics to assess and compare options for infrastructure

L.A. Shuster. What does infrastructure have to do with social justice and equity? Civil Engineering (Magazine) American Society of Civil Engineers: 2021.
Available: <https://www.asce.org/publications-and-news/civil-engineering-source/civil-engineering-magazine/issues/magazine-issue/article/2021/11/what-does-infrastructure-have-to-do-with-social-justice-and-equity>

S.S. Clark, S.K.E. Peterson, M.A. Shelly, R.F. Jeffers. Developing an equity-focused metric for quantifying the social burden of infrastructure disruptions. Sustainable and Resilient Infrastructure, 8(1). Taylor and Francis: 2023.
Available: <https://www.nrel.gov/docs/fy23osti/84940.pdf>